

```
java --version
sudo apt update
sudo apt install maven -y
```

```
mvn archetype:generate -DgroupId=com.example.app \
-DartifactId=demo-app \
-DarchetypeArtifactId=maven-archetype-quickstart \
-DinteractiveMode=false
```

```
cd demo-app
ls
nano pom.xml
```

POM.XML:

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-
v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example.app</groupId>
  <artifactId>demo-app</artifactId>
  <packaging>jar</packaging>
  <version>1.0-SNAPSHOT</version>
  <name>demo-app</name>
  <url>http://maven.apache.org</url>
  <dependencies>
    <dependency>
      <groupId>ch.qos.logback</groupId>
      <artifactId>logback-classic</artifactId>
      <version>1.4.14</version>
    </dependency>

    <!-- JUnit for testing -->
```

```
<dependency>
  <groupId>junit</groupId>
  <artifactId>junit</artifactId>
  <version>4.13.2</version>
  <scope>test</scope>
</dependency>
```

```
<!-- Logging -->
```

```
<dependency>
  <groupId>org.slf4j</groupId>
  <artifactId>slf4j-api</artifactId>
  <version>2.0.9</version>
</dependency>
```

```
<!-- Apache Commons -->
```

```
<dependency>
  <groupId>org.apache.commons</groupId>
  <artifactId>commons-lang3</artifactId>
  <version>3.14.0</version>
</dependency>
```

```
</dependencies>
```

```
<build>
```

```
<plugins>
```

```
<!-- Compiler Plugin -->
```

```
<plugin>
  <groupId>org.apache.maven.plugins</groupId>
  <artifactId>maven-compiler-plugin</artifactId>
  <version>3.11.0</version>
  <configuration>
```

```
<source>17</source>
<target>17</target>
</configuration>
</plugin>

<!-- Test Plugin -->
<plugin>
  <groupId>org.apache.maven.plugins</groupId>
  <artifactId>maven-surefire-plugin</artifactId>
  <version>3.1.2</version>
</plugin>

</plugins>
</build>
</project>
```

```
nano src/main/java/com/example/app/App.java
```

APP.JAVA:

```
package com.example.app;
```

```
import org.slf4j.Logger;
```

```
import org.slf4j.LoggerFactory;
```

```
import org.apache.commons.lang3.StringUtils;
```

```
public class App {
```

```
    private static final Logger logger = LoggerFactory.getLogger(App.class);
```

```
    public static void main(String[] args) {
```

```
        String name = "CI/CD Pipeline";
```

```
    if (StringUtils.isNotBlank(name)) {  
        String message = greet(name);  
        logger.info(message);  
        System.out.println(message);  
    } else {  
        logger.error("Name is empty!");  
    }  
}
```

```
public static String greet(String name) {  
    return "Hello, " + name + "! Welcome to Maven CI/CD Demo.";  
}  
}
```

nano src/test/java/com/example/app/AppTest.java

```
package com.example.app;
```

```
import org.junit.Test;
```

```
import static org.junit.Assert.assertEquals;
```

```
public class AppTest {
```

```
    @Test
```

```
    public void testGreet() {
```

```
        String result = App.greet("Student");
```

```
        assertEquals("Hello, Student! Welcome to Maven CI/CD Demo.", result);
```

```
    }
```

```
}
```

mvn clean install

```
mvn exec:java -Dexec.mainClass="com.example.app.App"
```

```
git config --global user.name "KhushiJigali"
```

```
git config --global user.email "khushikiranjigali@gmail.com"
```

```
git init
```

```
git add .
```

```
git commit -m "Initial Commit"
```

```
git remote add origin https://github.com/KhushiJigali/mavenp.git
```

```
git branch -M master
```

```
git push -u origin master
```

```
cd ..
```

```
sudo rm -rf ~/.jenkins
```

```
sudo rm jenkins.war
```

```
sudo apt update
```

```
sudo apt install openjdk-21-jdk -y
```

```
wget https://get.jenkins.io/war-stable/latest/jenkins.war
```

```
java -jar jenkins.war --httpPort=8080
```

Jenkinsfile:

```
pipeline {
```

```
    agent any
```

```
    tools {
```

```
        maven 'Maven'
```

```
        jdk 'JDK21'
```

```
    }
```

```
    stages {
```

```
        stage('Checkout') {
```

```
            steps {
```

```
        git branch: 'master', url:'https://github.com/KhushiJigali/maven-demo.git',
        credentialsId: 'github-token'
    }
}

stage('Build') {
    steps {
        sh 'mvn clean compile'
    }
}

stage('Test') {
    steps {
        sh 'mvn test'
    }
}

stage('Package') {
    steps {
        sh 'mvn package'
    }
}

stage('Run Application') {
    steps {
        sh 'mvn exec:java -Dexec.mainClass="com.example.app.App"'
    }
}
}
```

#EMAIL NOTIFICATION

```
pipeline {
  agent any

  tools {
    maven 'Maven'
    jdk 'JDK21'
  }

  stages {

    stage('Checkout') {
      steps {
        git branch: 'master',
            url: 'https://github.com/KhushiJigali/maven-email.git',
            credentialsId: 'github-token'
      }
    }

    stage('Build') {
      steps {
        sh 'mvn clean compile'
      }
    }

    stage('Test') {
      steps {
        sh 'mvn test'
      }
    }
  }
}
```

```
stage('Package') {  
    steps {  
        sh 'mvn package'  
    }  
}
```

```
stage('Run Application') {  
    steps {  
        sh 'mvn exec:java -Dexec.mainClass="com.example.app.App"'  
    }  
}
```

```
post {  
    success {  
        emailx (   
            subject: "SUCCESS: ${JOB_NAME} #${BUILD_NUMBER}",  
            body: "Build succeeded!\nCheck: ${BUILD_URL}",  
            to: "khushikiranjigali@gmail.com"  
        )  
    }  
}
```

```
failure {  
    emailx (   
        subject: "FAILED: ${JOB_NAME} #${BUILD_NUMBER}",  
        body: "Build failed!\nCheck: ${BUILD_URL}",  
        to: "khushikiranjigali@gmail.com"  
    )  
}  
}
```